

Trends in agricultural impact and recovery of wetlands in prairie Canada

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Abstract. Despite widespread recognition that they provide valuable ecosystem services and contribute significantly to global biodiversity, over half of the world's wetlands have been lost, primarily to agriculture. Wetland loss is evident in prairie Canada, but comprehensive information about causes of ongoing impact for existing wetlands is lacking. Habitat data collected for ~10 500 wetlands during annual waterfowl surveys (1985–2005) were analyzed using multistate models to estimate rates of wetland impact and recovery from agricultural activities in the Canadian prairies. An impact was defined as an agricultural activity that visibly altered a wetland margin (natural vegetation surrounding wetland interiors) or basin (interior depression capable of holding water), whereas recovery was deemed to have occurred if agricultural activities had ceased and effects were no longer visibly apparent. We estimated separate impact and recovery rates for wetland basins and wetland margins and considered covariates such as location, time, wetness indices, land use, and wetland permanence. Results indicate that impact rates for wetland margins have declined over time, likely due to a decreasing percentage of unaffected wetlands on the landscape. Recovery rates for margins were always lower than impact rates, suggesting progressive incidence of impacts to wetlands over time. Unlike margins, impact and recovery rates for basins fluctuated with May pond densities, which we used as a wetness index. Shallow ephemeral wetlands located in agricultural fields had the highest impact and lowest recovery rates relative to wetlands with higher water permanence or situated in areas of lower agricultural intensity. High rates and incidence of wetland impact in conjunction with low recovery rates clearly demonstrate the need for stronger wetland protection in prairie Canada.

Key words: agriculture; covariates; impact rates; multistate models; prairie Canada; recovery rates; wetlands.

INTRODUCTION

The ecological significance of wetlands is widely acknowledged, and global efforts have sought to prevent further degradation and destruction of wetland habitat (Mitsch and Gosselink 2000). Less than 9% of the earth's land area is covered by wetlands, but they support disproportionately high numbers of species (Zedler and Kercher 2005, Dudgeon et al. 2006) and provide habitat for waterbirds, amphibians, fish and other vertebrates, invertebrates, and aquatic and terrestrial flora (Mitsch and Gosselink 2000). In addition, wetlands supply ecosystem services such as water quality improvement, flood control, nutrient cycling, and carbon sequestration (Zedler and Kercher 2005, Daniels and Cumming 2008). Despite acknowledgment of their

substantial ecological, cultural, and economic contributions, over half of the world's wetlands have been lost (Zedler and Kercher 2005). Losses have been primarily attributed to agriculture, while many remaining wetlands have been modified through hydrological alteration, salinization, eutrophication, sedimentation, filling, and/or invasion of exotic species (Zedler and Kercher 2005). These activities can produce persistent but subtle effects, and thus are more difficult to quantify than obvious declines in wetland numbers. Moreover, impacts may not be permanent; depending on their severity, wetlands may recover to pre-impact conditions given time and/or restoration efforts (Brock et al. 2003, Lindig-Cisneros et al. 2003). Consequently, estimates of wetland impacts are not straightforward or readily available for many areas.

To effectively plan and deliver wetland restoration and protection programs, determining whether and how quickly wetlands can recover from impacts is critical (Lindig-Cisneros et al. 2003, Heffernan 2008). Understanding how putative factors contribute to wetland impact and recovery can help identify wetlands at risk

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and, thereby, focus conservation efforts where they are most needed and effective. Significant variation in wetland impact and recovery rates may arise as a result of variability in: (1) anthropogenic factors, such as land uses and economies; (2) climatic factors, such as temperature and precipitation; (3) conditions specific to individual wetlands, such as depth and wetland type; and/or (4) other environmental factors, such as local wetland density and vegetation (Bradshaw and Smit 1997, Johnson et al. 2005, Daniels and Cumming 2008). Effects of these factors on wetlands are likely numerous and complex, but few studies have attempted to parse out the contributions of these factors on wetland impact and recovery.

Wetlands of the North American prairie pothole region are landscape features undergoing continuous change through natural and anthropogenic processes (Mitsch and Gosselink 2000). The region is characterized by strong north-south temperature and east-west precipitation gradients, but with spatial heterogeneity, temporal instability, and most areas subject to periodic drought and deluge conditions (Johnson et al. 1994, Johnson et al. 2005). Before European settlement, large expanses of native grasslands were common, but fertile soils and accessibility made the prairie pothole region ideal for agriculture. As a result, most land has been converted to cropland or is managed as hay or pasture land (Johnson et al. 1994, Horn et al. 2005). To expand the agricultural land base, many wetlands have been drained, filled, and cultivated, resulting in an estimated 40–70% reduction in the number of wetlands on the landscape (Whigham 1999, Dahl 2000).

Wetland loss in the prairie pothole region continued in the last quarter of the twentieth century, but information on rates and incidence of agricultural impacts to existing wetlands for that same time period is limited (Dahl and Watmough 2007). In Canada, impact trends are uncertain as no comprehensive inventory or periodic monitoring program exists for wetlands (Dahl and Watmough 2007). However, since 1980, some wetland habitat data have been collected during the annual Waterfowl Breeding Population and Habitat Surveys (U.S. Fish and Wildlife Service and Canadian Wildlife Service 1987). Based on an analysis of these data, our objectives were to: (1) assess spatiotemporal variation in wetland impact rates across the Canadian prairie pothole region, and (2) determine how factors such as precipitation, land use, and wetland permanence may influence those rates. We conducted a broadscale, yet comprehensive, assessment of trends in wetland impacts, and evaluated predictions of how wetland “states” (i.e., agricultural impacts to margins and basins; see *Hypotheses and predictions* for definitions) were related to characteristics of flooding regimes, geographic region, and land use. We employed multi-state estimation techniques to accomplish these objectives (Brownie et al. 1993) in what we believe is the first

study to apply these techniques to quantify rates associated with wetland impact and recovery.

Hypotheses and predictions

We defined a wetland impact as any agriculture-related alteration to wetlands (e.g., grazing or cultivation). Wetland alterations may benefit certain species but are far more likely to have negative consequences for at least some native species (e.g., Lindeman and Clark 1999, Johnson and Dechant-Shaffer 2002, Gleason et al. 2003), and these changes may alter wetland community structure. Therefore, we assumed wetlands with impacts represented degraded habitat, regardless of potential benefits to particular species or groups of organisms.

During drought, wetland basins (i.e., the area extending from the center of the wetland to the outer edge of the wet meadow zone [Stewart and Kantrud 1971]) become exposed and vulnerable to agricultural impacts, but activities in affected basins cease when they re-flood (Whigham 1999). Therefore, we predicted rates of impact to wetland basins would be lower when spring seasons are wet and greater when basins are drawn down or dry. Because wetland margins (i.e., the area of native vegetation extending 10 m from the contour of the basin edge) are less prone to flooding than basins, we predicted margins would have a higher incidence of impacts than basins, and rates would be less affected by varying water conditions. Turner et al. (1987) reported wetlands with margin impacts progressively increased from 1981 to 1985, and we expected this trend continued after 1985. Although native land conversion to agriculture has slowed, activities on existing farmlands have intensified through conversion of remnant marginal lands, “improvement” of pasture land to support larger cattle herds, and diminished use of summer fallow, a practice of leaving land uncropped for an entire growing season (Bethke and Nudds 1995, Carlyle 1997, Beaulieu and Bedard 2003). Due to this agricultural intensification, we expected increasing agricultural pressure on wetland margins and less opportunity for recovery.

Our study area encompassed two major ecoregions: grassland and parkland (see *Methods*). Because water conditions tend to be more variable in grassland regions (Nudds 1983), we expected the incidence of impacts would generally be highest in grassland, and associated transition rates and percentages of affected basins would fluctuate more than in parkland. Moreover, wood-rings surrounding parkland wetlands could provide a buffer against agricultural activities such as grazing and cultivation when parkland basins are drawn down. We also anticipated permanent wetlands, being less exposed to agricultural practices due to the presence of water, would have a lower impact rate when compared with ephemeral or seasonal wetlands (Whigham 1999). Likewise, wetlands in cultivated fields or pastures were expected to have higher impact rates than wetlands located in idle grasslands or wooded areas. Finally, we expected jurisdictional differences in rates and incidence

of impact to wetlands due to provincial differences in agricultural and wetland protection policies as well as other socioeconomic drivers (Douglas and Johnson 1994, Bradshaw and Smit 1997).

METHODS

The Canadian portion of the prairie pothole region encompasses ~480 000 km² extending north from the Canada/United States border to the southern edge of the boreal forest (Fig. 1). The Canadian prairie pothole region is composed of several ecoregions (i.e., zones characterized by distinct biotic and abiotic factors including climate, physiography, vegetation, soil, water, and fauna [Marshall et al. 1999]), but was classified into two zones to simplify analysis and reporting: (1) grassland in southern Alberta and southwestern Saskatchewan, and (2) parkland and boreal transition (hereafter referred to as parkland) in central Alberta and Saskatchewan, eastern Saskatchewan, and the entire prairie portion of Manitoba. We anticipated landscape processes and impacts to wetlands would be similar within these broad ecoregions; further benefits of this classification included facilitation of comparisons with previous studies, adequate sample coverage at these scales, and avoiding constraints in computing capabilities when analyzing data collected at finer spatial resolutions.

The Waterfowl Breeding Population and Habitat Survey has been conducted annually by the U.S. Fish and Wildlife Service and Canadian Wildlife Service since 1955 (U.S. Fish and Wildlife Service and Canadian Wildlife Service 1987). To estimate sizes of duck populations, aerial and ground surveys are conducted each May over major breeding areas in Canada and the United States. These breeding areas are delineated into survey strata according to political boundaries, habitat conditions, and duck densities. Ground crews also visually assess habitat conditions at each wetland visited during surveys. All wetlands within survey segment boundaries have been assigned unique numbers and are labeled on aerial photos, so observers revisit every wetland annually (although in rare cases, wetlands are missed because of logistical issues). Consequently, a complete history of spring habitat conditions exists for individual wetlands within ground segments.

Each year during ground surveys, six major habitat variables are recorded at each wetland: (1) basin impacts and conditions, (2) margin impacts and conditions, (3) wetland type, (4) water level stage, (5) upland margin width, and (6) adjacent land uses (Turner et al. 1987). Impacts and conditions of basins and margins are identified and expressed as percentages of the area affected by haying, draining, clearing, grazing, burning, cultivation, filling, road building, farmyard impacts, and wooded area (Turner et al. 1987). Wetland type is related to water permanency and follows a classification scheme developed, in part, by Stewart and Kantrud (1971); artificial wetlands, such as dugouts and irrigation

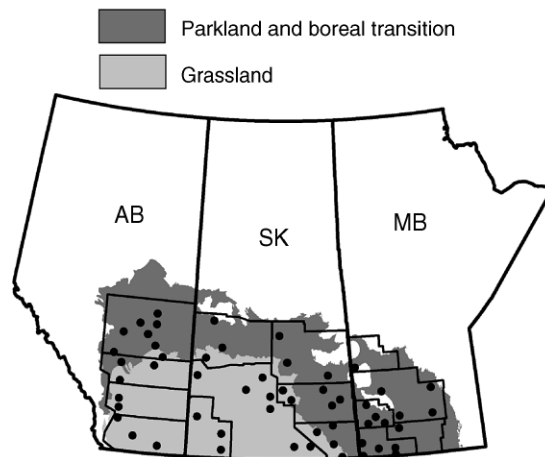


FIG. 1. Map of the parkland and boreal transition and the grassland ecoregions of the Canadian prairie pothole region. Boundaries of prairie air-ground survey strata are represented by the polygons within the provinces, and solid circles signify the locations of survey segments ($n = 56$). Province abbreviations are: AB, Alberta; SK, Saskatchewan; MB, Manitoba.

canals, are also recorded. Water level stage, a numerical index describing water levels relative to “normal” conditions and ranging from dry to overflowing, is assigned to each wetland. In addition, margin widths are estimated visually (nearest 1 m) with widths ≥ 10 m fixed at 10 m. Upland conditions adjacent to the outer edge of the margin are expressed as percentages of the perimeter of a wetland margin influenced by crop, stubble, fallow, hay, native grass, wood, pasture, wooded pasture, and farmyard. Of these habitat variables, we used impacts and conditions of basins and margins to determine the extent of impact for wetlands.

Survey protocol requires the previous year’s habitat data to be edited on-site during surveys to reduce observer bias and ensure only real habitat changes are recorded (Turner et al. 1987). Because wetland and upland assessments may be subjective, we quantified inter-observer variation in habitat assessments during the 2006 and 2007 Saskatchewan surveys. Each day when duck surveys were completed, crews returned to the last 2–4 wet ponds on segments and a new observer reassessed habitat conditions, using the same methods as the original observer. The types and number of discrepancies between observers were recorded and then expressed as percentages of the total wetland revisits each year. We first determined between-observer discrepancies in detecting impacts; this is relevant to multistate modeling that requires correct designation of a wetland’s “state” of impact (i.e., impact vs. no impact). If both observers detected impacts, we then calculated the percentage of misclassification of types of impacts between observers.

To examine variation in rates of impact and recovery for wetlands, we used multistate models implemented in Program MARK (White and Burnham 1999, White

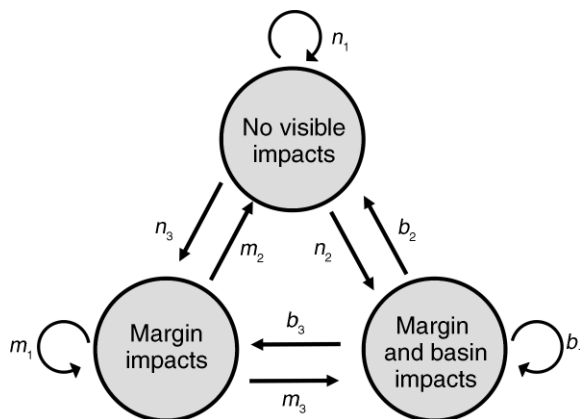


FIG. 2. Multistate framework depicting possible wetland states and transitions that can occur between years: n_1 = probability a wetland with no visible impacts will remain in that state; n_2 = probability a wetland with no visible impacts will shift to a wetland with margin and basin impacts; n_3 = probability a wetland with no visible impacts will shift to a wetland with margin impacts; m_1 = probability a wetland with margin impacts will remain in that state; m_2 = probability a wetland with margin impacts will revert to a wetland with no visible impacts; m_3 = probability a wetland with margin impacts will shift to a wetland with margin and basin impacts; b_1 = probability a wetland with margin and basin impacts will remain in that state; b_2 = probability a wetland with margin and basin impacts will revert to a wetland with no visible impacts; b_3 = probability a wetland with margin and basin impacts will revert to wetland with margin impacts.

et al. 2006). Program MARK provides a suite of mark-recapture models used to estimate survival and recapture/recovery parameters for marked populations (White and Burnham 1999). Basic input data for analyses are encounter histories reflecting the sequence of between-year recaptures or resightings of marked individuals. Survival and recapture parameters are computed by maximum likelihood estimation, and parameter estimates can be modeled in relation to time-varying and individual covariates, which can help elucidate effects of explanatory variables (White and Burnham 1999).

Multistate models are extensions of standard mark-recapture models (i.e., Cormack-Jolly-Seber models; Lebreton et al. 1992), which, in addition to survival and recapture probabilities, can estimate transition probabilities (Ψ) between states (Brownie et al. 1993, White et al. 2006). States can represent physical locations where Ψ signifies movement probabilities, or they can represent conditions where Ψ signifies probabilities of change between conditions (Nichols et al. 1994). Although multistate models were originally developed to quantify movement and other state transitions among marked individuals, they also provide an avenue for estimating rates of change associated with dynamic landscape processes such as wetland impact and recovery. Multistate models may be ideally suited for such processes because they can simultaneously

estimate several probabilities, and several spatiotemporal factors can be incorporated to determine how probabilities covary with environmental and anthropogenic conditions.

We considered wetlands as marked individuals and used multistate models to estimate rates of transition among three impact states (Fig. 2). Complete destruction occurred for <1% of all surveyed wetlands over the entire study period (1985–2005); therefore, we set survival probability to 1 (White and Burnham 1999). Because an objective of the ground survey is to visit every wetland every year, we selected wetlands with complete records for all 21 occasions (total of 10 437 encounter histories for individual wetlands from 56 survey segments; Fig. 1). Exclusive use of wetlands visited each year of study allowed us to also fix recapture probability at 1 (White and Burnham 1999). As a result, model development and parameter estimation was based solely on transition probabilities, which are synonymous with rates of change and reflect the probability that an individual in state i will shift to state j between two sampling occasions (i.e., years). Because impact rates may vary between wetland margins and basins, we considered three possible wetland states (Fig. 2): (1) no visible impact—no impacts recorded (hereafter referred to as pristine); (2) margin impact—an impact recorded only for a wetland's margin (hereafter referred to as partial impact); and (3) margin and basin impact—an impact recorded both for a wetland's margin and basin (hereafter referred to as full impact). Between years, wetlands could remain in their current state or shift to another state. Probabilities of transition of wetlands to states with greater impact are synonymous with impact rates, and probabilities of transition to less-affected states represent recovery rates. The term "recovery" as used here refers to the probability of a wetland reverting to a less-affected state, and should not be confused with recovery as it applies to conventional mark-encounter studies (where "recovery probability" is the probability that a marked individual is killed, retrieved, and reported by a hunter; Brownie et al. 1985).

We considered spatiotemporal effects as well as characteristics of individual wetlands as sources of variation in transition probabilities. We defined five attribute groups based on provincial and regional locations: Alberta grassland and parkland; Saskatchewan grassland and parkland; and Manitoba parkland. Survey strata were classified according to these attribute groups based on their provincial location and ecoregion that made up the majority of each stratum's area (Fig. 1). We calculated two annually varying wetness indices at the level of attribute group, and analyzed these as cohort-level covariates (White and Burnham 1999). The first index, pond density, was calculated from historical May pond estimates per stratum and stratum areas provided by the Waterfowl Breeding Population and Habitat Survey (Wilkins et al. 2006). After determining ponds per square kilometer for each stratum, we calculated an

average pond density weighted by stratum areas for each attribute group. The second index, conserved soil moisture, was developed by Williams and Robertson (1965) to aid in forecasting prairie wheat production on the basis of precipitation data. Conserved soil moisture (CSM) is a form of weighted mean of monthly precipitation for the 21 months prior to May of year t :

$$\begin{aligned} \text{CSM} = & 0.36A + [0.37B - 0.2(0.36A)] + 0.13C \\ & + \{0.30D - 0.2[0.36A + 0.37B - 0.2(0.36A) \\ & + 0.13C]\} \end{aligned}$$

where A = total precipitation from August through October in year $t - 2$, B = total precipitation during November in year $t - 2$ through April in year $t - 1$, C = total precipitation during May through October in year $t - 1$, and D = total precipitation during November in year $t - 1$ to April in year t . Conserved soil moisture has been used as an index for wetness regimes and wetland conditions, in some instances with stronger correlations to duck abundance than pond numbers (Boyd 1981, Bethke and Nudds 1995). To calculate annual conserved soil moisture for each attribute group, we used monthly precipitation values averaged over all Environment Canada (2007) weather stations operating from 1983 to 2005. We substituted each of the two calculated wetness indices in the models to determine which index performed best in explaining variation in wetland impact rates. In some models, we constrained transition rates to vary as linear-logistic functions of time (calendar year; T) as this can be useful for identifying systematic changes (i.e., temporal trends) in transition rates.

In addition to assessing spatial and temporal variation in impact and recovery rates, we also included dominant wetland type (POND) and land use (LAND) as covariates. Although these factors can vary over time and Program MARK allows for time-varying individual covariates, we assigned a single dominant classification due to computing-related constraints on model complexity. Each wetland was assigned a dominant wetland type based on its type designation at full water stage, as this was the stage at which we had the greatest confidence in wetlands being correctly classified. If a wetland was typed differently at full stages on different occasions, the type assigned on a majority of visits was used. When different type designations occurred with equal frequency, wetlands were assigned the more permanent type. If wetlands were never observed at full water level stages, they were assigned a dominant wetland type based on the highest water level stage encountered. For purposes of analysis, Type 1 (temporary) and Type 3 (seasonal) wetlands were combined as seasonal wetlands, and Type 4 (semipermanent) and Type 5 (permanent) were combined as permanent wetlands. Artificial ponds (e.g., dugouts, gravel pits, irrigation canals) and streams were given separate respective classes from other wetlands. Finally, wetlands never holding enough water to last more than a few days

were classified separately as ephemeral wetlands (Brock et al. 2003). Each wetland was assigned a dominant land use based on most frequently observed upland condition. Categories included pasture, hayed/yard, cultivated, wood, native grass, and mixed in cases where different land uses occurred with equal frequency over time.

We used a two-stage approach for candidate model set development. First, we assessed which spatial factor (province, ecoregion, or province and ecoregion combined) was best supported by the data. Based on results of this initial analysis, we selected the most appropriate spatial factor for inclusion in all subsequent models testing temporal and individual covariates. Based on a priori predictions, we developed a candidate set of models that included all possible additive combinations of the most supported spatial factor and the aforementioned covariates (15 models), a null model, and additive and multiplicative models allowing for both group and time dependence (group + t , group * t , respectively). Model notation follows Lebreton et al. (1992), where “+” indicates an additive relationship between explanatory variables, and “*” indicates an interactive relationship. We used a logit link function to model the relationships between transition parameters and covariates.

As a preliminary step, we assessed overall goodness of fit using the median- $\hat{c}$ test in Program MARK (White and Burnham 1999). Because median- $\hat{c}$ goodness-of-fit testing does not allow inclusion of individual covariates, we performed the test on a highly parameterized non-global model, group + t (Burnham and Anderson 2002). The median- $\hat{c}$ test had a computer run-time of three weeks, which is why we did not run the median- $\hat{c}$ test on the most parameterized model, group * t . We also adjusted the variance inflation factor ($\hat{c}$) from 1 to 4 in computing model diagnostics to observe effects on model rankings (QAIC_c values; Burnham and Anderson 2002).

Model selection was based on an information-theoretic approach by computing Akaike's Information Criterion adjusted for sample size and overdispersion (QAIC_c; Burnham and Anderson 2002). QAIC_c indicates a model's degree of parsimony, which represents a balance between model fit and precision of parameter estimates. Consequently, models within each candidate set can be ranked based on their QAIC_c values, with the lowest QAIC_c indicating the most parsimonious model. To aid in model selection, we also computed ΔQAIC_c (QAIC_c values expressed relative to QAIC_c of the most parsimonious model) and AIC-based model weights (Burnham and Anderson 2002). After identifying the most parsimonious additive model, we ran a multiplicative equivalent that included an interaction between spatial factors and cohort-level covariates. We report transition parameter estimates and standard errors from the best approximating model, and also examine percentages of affected wetland margins and basins over

TABLE 1. Multistate model selection results estimating transition probabilities for wetland impacts across the Canadian prairie pothole region (1985–2005).

Model	QAIC _c	ΔQAIC _c	QAIC _c weights	No. parameters
Group	78 220.98	0.00	1.00	30
Provinces	78 362.33	141.35	0.00	18
Ecoregions	79 493.20	1272.22	0.00	12
Null	79 635.36	1414.38	0.00	6

Notes: The candidate set was used to determine how probabilities varied spatially with survival and recapture probabilities fixed at 1. Akaike's Information Criterion was corrected for sample size and adjusted for overdispersion ($\hat{c} = 1.72$). Spatial models included: (1) no spatial variation (null); (2) ecoregional variation—grassland and parkland (ecoregions); (3) provincial variation—Alberta, Manitoba, and Saskatchewan (provinces); and (4) a combination of province and ecoregion (group).

the study period for validation and interpretation of the multistate model results.

RESULTS

In 2006, 93% and 91% of margin and basin impacts, respectively, were consistently detected by two observers; in 2007, corresponding estimates were 100% and 97%. These values are especially relevant to the multistate modeling exercise because detection of impacts determines a wetland's state; detection failures were also considered when interpreting results.

Using the median- $\hat{c}$ method, the variance inflation factor ($\hat{c}$) was estimated at 1.72 for the model that contained additive effects of group variation and time; adjusting $\hat{c}$ from 1 to 4 did not change the ΔQAIC_c ranking or QAIC_c weight of the most parsimonious model.

Of the spatial factors considered, the model allowing for both provincial and ecoregional variation (i.e., group) was most supported (QAIC_c weight = 1.00; Table 1). We subsequently included this grouping factor in all models testing for effects of individual and time-varying covariates (Table 2). As models including pond density always received more support than conserved soil moisture equivalents, we included pond density in the following combined-effects model. Because we predicted different transition parameters would respond differently to specific covariates, we developed a model (group + T/PD + LAND + POND) that constrained three transition parameters (m_3 , b_3 , and n_2) to vary as a function of pond density and three transition parameters (m_2 , b_2 , and n_3) to vary as a function of T (Fig. 2). Of the additive models, this combined-effects model received the most support, and its multiplicative equivalent (group * T/PD + LAND + POND) was most parsimonious (QAIC_c weight = 1.00; Table 2). Consequently, we report transition parameter estimates from this multiplicative model.

To illustrate spatial and temporal patterns of variation in transition rates (group * T/PD), we present estimates corresponding to seasonal ponds in cultivated landscapes, as these were the most common wetland type and land use, respectively. Because land use and wetland type were additive effects in the best approximating model, the temporal trends are the same for all land uses and wetland types, although the magnitudes of estimates may vary. At the beginning of the study period, group differences in rates of transition from pristine to partially impacted wetlands were evident (Fig. 3): Alberta had the lowest impact rates at 0.12 ± 0.04 (estimate $\pm$ SE) for grassland and 0.11 ± 0.03 for parkland, Manitoba had the highest rate at 0.26 ± 0.02 , and Saskatchewan

TABLE 2. Multistate model selection results for estimating transition probabilities of wetlands across the Canadian prairie pothole region (1985–2005).

Model†	QAIC _c	ΔQAIC _c	QAIC _c weights	No. parameters
group * T/PD + LAND + POND	72 747.06	0.00	1.00	114
MPD * T/PD + LAND + POND‡	72 858.21	111.15	0.00	96
group + T/PD + LAND + POND	72 901.84	154.78	0.00	90
group + T + LAND + POND	73 362.36	615.29	0.00	90
group + PD + LAND + POND	73 573.34	826.28	0.00	90
group + T + POND	74 260.25	1513.18	0.00	60
group + CSM + LAND + POND	74 303.42	1556.36	0.00	90
group * t	74 453.06	1705.99	0.00	600
group + PD + POND	74 525.88	1778.81	0.00	60
group + LAND + POND	74 597.38	1850.32	0.00	84

Notes: Survival and recapture probabilities were fixed at 1. Results from the top-ranked 10 candidate models are displayed. Akaike's Information Criterion results were corrected for sample size and adjusted for overdispersion ($\hat{c} = 1.72$).

† Model covariates and symbols: group = combination of provinces and ecoregions; MPD = maximum wetland basin density (constant); t = time dependency (i.e., year-specific variation); T = linear-logistic trend over time; PD = annually varying pond density at group level; CSM = annually varying conserved soil moisture at group level; T/PD = three transition parameters constrained linearly with time, and three transition parameters constrained by pond density; LAND = dominant land use (individual covariate); POND = dominant wetland type (individual covariate); "+" indicates additive effects; "*" indicates multiplicative effects.

‡ A posteriori model.

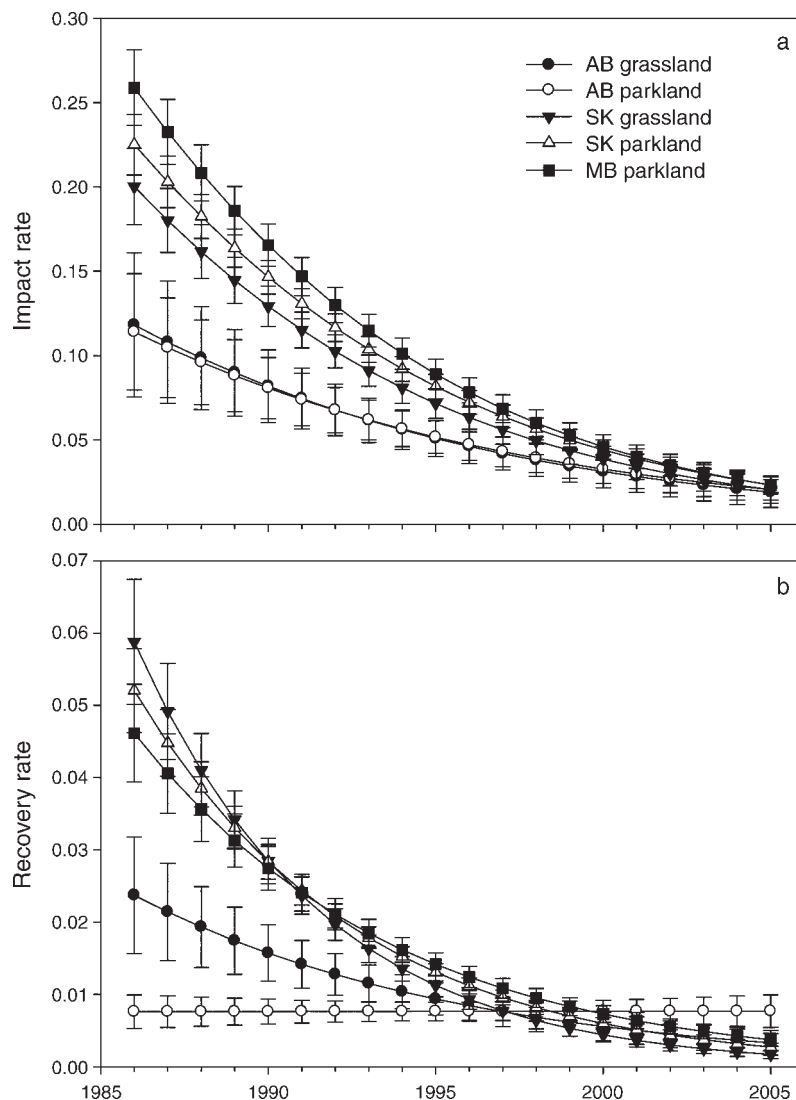


FIG. 3. Estimates of transition probabilities ($\pm$ SE) derived from model (group * $T/$ PD + LAND + POND) for seasonal wetlands in cultivated landscapes, 1985–2005. Refer to Table 2 for definitions of model covariates. (a) Rates of transition from no visible impacts to wetlands with margin impacts. (b) Rates of transition from margin impacts to wetlands with no visible impacts. Note that the y-axis scales differ. Province abbreviations are: AB, Alberta; SK, Saskatchewan; MB, Manitoba.

grassland and parkland regions were intermediate at 0.20 ± 0.02 and 0.23 ± 0.02 , respectively. Rates declined for all groups over time and converged on a similar value near 0.02 in 2005. The reverse transition probabilities, recovery rates, displayed similar declining trends over time. However, magnitudes of recovery rates were much lower, implying progressive incidence of impacts to wetlands over the study period (Fig. 3).

The most parsimonious model constrained transition probabilities between partially and fully impacted wetlands to vary as a function of pond densities (Fig. 4). In general, rates of impact on wetland basins declined as pond densities increased. Conversely, recovery rates increased with increasing pond density, but at lower magnitudes than impact rates. Alberta grassland was the

exception and displayed a slight decline in recovery rates with increasing pond density. Where pond densities overlapped for all groups, Saskatchewan grassland and parkland had the highest impact rates, Manitoba was mid-range, and Alberta parkland and grassland rates were the lowest (Fig. 4).

Rates of transition from pristine to fully impacted wetlands were constrained to vary by pond density, and corresponding recovery rates were constrained to vary as a linear-logistic function of time (Fig. 5). Impact rates declined similarly (slopes close to parallel) in all groups with an increase in pond density, but occurred at different magnitudes (intercepts) along provincial divisions. Saskatchewan and Alberta wetland basins had the highest and lowest impact rates, respectively. Recovery

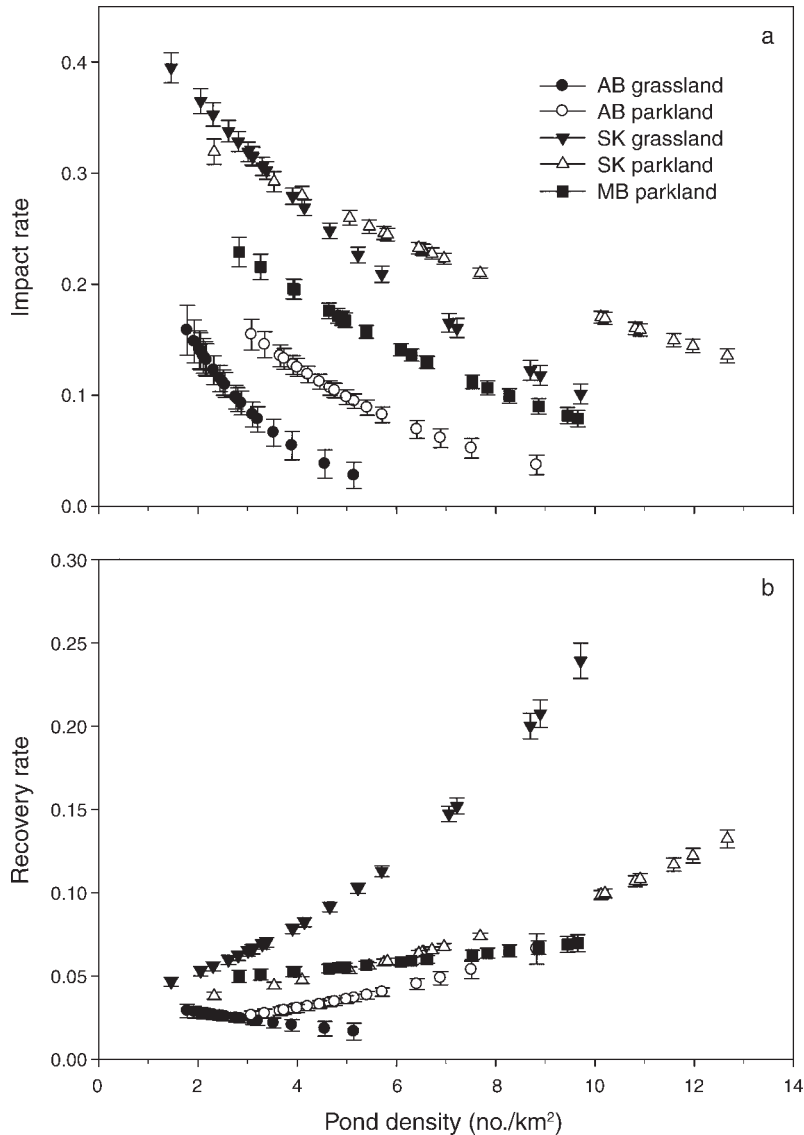


FIG. 4. Estimates of transition probabilities ($\pm$ SE) derived from model (group * $T/PD + LAND + POND$) for seasonal wetlands in cultivated landscapes relative to pond density, 1985–2005. Refer to Table 2 for definitions of model covariates. Pond density was calculated from May pond estimates in the Waterfowl Breeding Population and Habitat Survey. (a) Rates of transition from margin impacts to wetlands with basin and margin impacts. (b) Rates of transition from basin and margin impacts to wetlands with margin impacts only. Note that the y-axis scales differ. Province abbreviations are: AB, Alberta; SK, Saskatchewan; MB, Manitoba.

rates were low (≤ 0.01) and declined through time for all groups, except Alberta parkland where recovery rates remained constant throughout the study period (Fig. 5).

To determine whether variation in wetland impacts between attribute groups was merely a function of landscape-specific differences in wetland densities, we considered an additional a posteriori model. As a proxy for total wetland basin density, we substituted each group in the best-supported a priori model with its maximum pond density (MPD) recorded during the study period ($MPD * T/PD + LAND + POND$). If this a posteriori model received more support than the a priori

equivalent with group variation, then we may have concluded that the observed group variation was a function of differences in the densities of total wetland basins in each geographic region. However, the a posteriori model was ranked second in the candidate set and did not receive any measurable support ($QAIC_c$ weight = 0.00; Table 2).

We chose seasonal wetlands in Saskatchewan parkland from 1995 to illustrate land-use effects because these wetlands comprised the majority of sampled wetlands at the midpoint of the study period. Generally, wetlands in cultivated or pasture lands had the highest

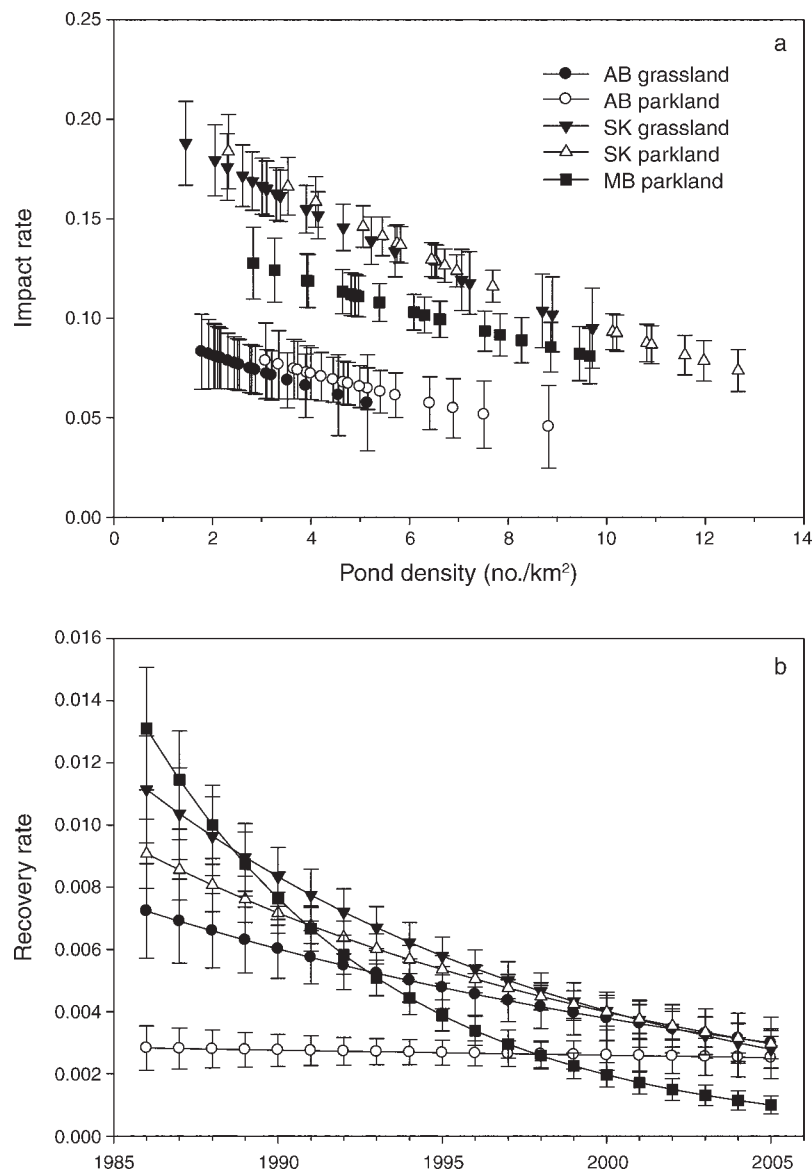


FIG. 5. Estimates of transition probabilities ($\pm$ SE) derived from model (group * $T/PD + LAND + POND$) for seasonal wetlands in cultivated landscapes, 1985–2005. Refer to Table 2 for definitions of model covariates. Pond density was calculated from May pond estimates in the Waterfowl Breeding Population and Habitat Survey. (a) Rates of transition from no visible impacts to wetlands with basin and margin impacts relative to pond density. (b) Rates of transition from basin and margin impacts to wetlands with no visible impacts constrained to vary as a linear-logistic function of time. Note that the x- and y-axis scales differ. Province abbreviations are: AB, Alberta; SK, Saskatchewan; MB, Manitoba.

impact and lowest recovery rates (Fig. 6a,c). In contrast, wetlands in native grassland and wooded areas had the lowest impact and highest recovery rates. Exceptions were transition probabilities between partially and fully impacted wetlands (Fig. 6b), where impact rates and corresponding recovery rates were close to equal for all land uses. Impact rates were still highest in cultivated and pasture lands, but their recovery rates were almost as equally high.

As with land-use effects, we chose Saskatchewan parkland in 1995 to illustrate effects of wetland type,

and we chose wetlands with adjacent cultivation because this was the most common land use. Impact rates were higher than recovery rates for all wetland types for the transitions from pristine to partially impacted wetlands (n_3 and m_2) and pristine to fully impacted wetlands (n_2 and b_2) (Table 3). Artificial and permanent wetlands had the highest impact rates for the transition from pristine to partially impacted; ephemeral wetlands had the highest impact rates for the transition from pristine to fully impacted (Table 3). Seasonal and permanent wetlands had similar impact and recovery rates for

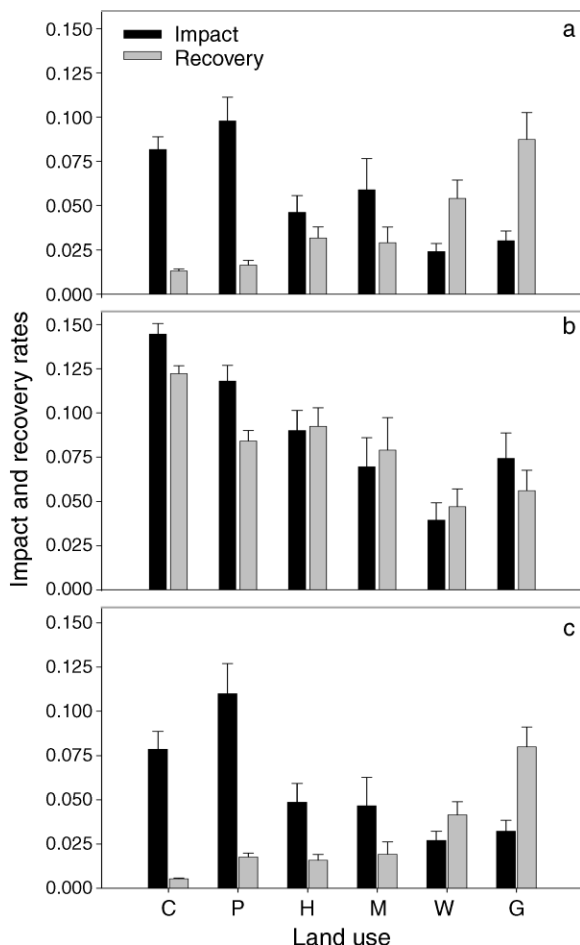


FIG. 6. Estimates of transition probabilities ($\pm$ SE) derived from model (group $\times$ T/PD + LAND + POND) for wetlands surrounded by different land uses in Saskatchewan parkland, 1995. (a) Impact and recovery rates are estimates of transition probabilities between wetlands with no visible impacts and wetlands with margin impacts. (b) Impact and recovery rates are estimates of transition probabilities between wetlands with margin impacts and wetlands with basin and margin impacts. (c) Impact and recovery rates are estimates of transition probabilities of wetlands with no visible impacts and wetlands with basin and margin impacts. Land-use categories are: C, cultivation; P, pasture; H, hay/farmyard; M, mixed use; W, woodland; G, grassland (native and restored).

transitions between pristine and partially impacted states, but seasonal wetlands had a higher impact rate for pristine to fully impacted. For the transitions between partially and fully impacted wetlands (m_3 and b_3), ephemeral wetlands had the highest impact rate but lowest recovery rate, and artificial wetlands had the lowest impact rate and highest recovery rate (Table 3). Seasonal wetlands had a higher impact rate than permanent wetlands, but recovery rates were relatively similar between the two types (Table 3).

We also used survey data to examine spatiotemporal variation in frequency of impacts and corroborate inferences made from multistate model results. In

1985, the percentage of wetlands with margin impacts ranged from 82% in Manitoba to 97% in Alberta grassland (Fig. 7). Until the early 1990s, the percentage of affected margins increased in Saskatchewan and Manitoba, but declined in Alberta. From 1992 onward, margin impact percentages remained relatively stable in all areas. In 2005, Alberta grassland and Manitoba had the lowest margin impact percentage at 92%; Saskatchewan grassland was highest with 95%.

Directional trends in the percentage of wetlands with basin impacts were not as clear as the trends for margin impacts. Proportions fluctuated in both Saskatchewan and Manitoba, but Alberta displayed a slight downward trend for the first half of the study period before leveling off (Fig. 7). In general, percentages of basin impacts were less than margin impacts, ranging from as low as 55% in Manitoba to 86% in Alberta grassland.

DISCUSSION

Until now, quantitative information on trends and status of Canadian prairie wetlands has largely been lacking. Some evidence suggests wetland loss has continued in the Canadian prairies, but trends in impacts have remained largely unknown due to no comprehensive inventory or periodic monitoring program (Dahl and Watmough 2007). Results from this study clearly demonstrate high rates of wetland impact across the entire Canadian prairies. Although impact rates on pristine wetland margins differed among provinces and ecoregions at the beginning of the study, they converged on similarly low rates by the mid-2000s after wetlands had attained a high percentage of impact. Impacts to wetland basins fluctuated with precipitation throughout the study period, with ephemeral wetlands being the most vulnerable to impact and slowest to recover. Results also indicate adjacent land uses influenced wetland impact rates.

The most parsimonious models we used included effects of province and ecoregion, linear trends over time, pond density, land use, and wetland type to explain spatial and temporal variation in wetland impact rates. In general, trends in estimates of transition parameters from the best approximating model were consistent with initial predictions. At the beginning of the study, incidence of wetland impacts was highest in Alberta, but impact rates were higher in Manitoba and Saskatchewan. Given that impact rates on wetland margins were higher than recovery rates, we expected an increased incidence of wetlands with margin impacts over time. While this was found to be the case for Manitoba and Saskatchewan, the incidence of margin impacts in Alberta was initially high and subsequently declined to levels similar to those seen in the other prairie provinces. Although recovery rates were lower than impact rates in Alberta, the high percentage of already-impacted margins meant more wetlands returned to a pristine state than were modified to a partially impacted state. For all groups, impact rates on

TABLE 3. Estimates of transition probabilities ($\pm$ SE) for different wetland types in a cultivated landscape in Saskatchewan parkland, 1995.

Wetland type	n_3	m_2	m_3	b_3	n_2	b_2
Ephemeral	0.04 ± 0.01	0.01 ± 0.00	0.31 ± 0.02	0.02 ± 0.00	0.14 ± 0.02	0.00 ± 0.00
Seasonal	0.08 ± 0.01	0.01 ± 0.00	0.14 ± 0.01	0.12 ± 0.00	0.08 ± 0.01	0.01 ± 0.00
Permanent	0.09 ± 0.01	0.02 ± 0.00	0.05 ± 0.00	0.11 ± 0.01	0.05 ± 0.01	0.00 ± 0.00
Artificial	0.09 ± 0.01	0.02 ± 0.00	0.02 ± 0.00	0.19 ± 0.02	0.02 ± 0.01	0.01 ± 0.00
Stream	0.07 ± 0.01	0.02 ± 0.00	0.07 ± 0.01	0.14 ± 0.01	0.04 ± 0.01	0.01 ± 0.00

Notes: Estimates are derived from model (group * T/PD + LAND + POND); n_3 = rates of transition from wetlands with no visible impacts (pristine) to wetlands with margin impacts (partially impacted); m_2 = rates of transition from wetlands with margin impacts to wetlands with no visible impacts; m_3 = rates of transition from wetlands with margin impacts to wetlands with basin and margin impacts (fully impacted); b_3 = rates of transition from wetlands with basin and margin impacts to wetlands with margin impacts; n_2 = rates of transition from wetlands with no visible impacts to wetlands with basin and margin impacts; b_2 = rates of transition from wetlands with basin and margin impacts to wetlands with no visible impacts.

pristine wetland margins declined with time, and percentages of wetlands with affected margins stabilized between 90% and 95% in the early 1990s and remained at those levels until the end of this study.

We hypothesized that basin impacts would occur more frequently during drought conditions, and esti-

mates of impact and recovery rates clearly demonstrate that basin impact rates are a function of water conditions in spring. When the spring season was wet and pond densities were high, basin impact rates were low and recovery rates were high; the converse was true in dry springs (i.e., low pond densities). The exception

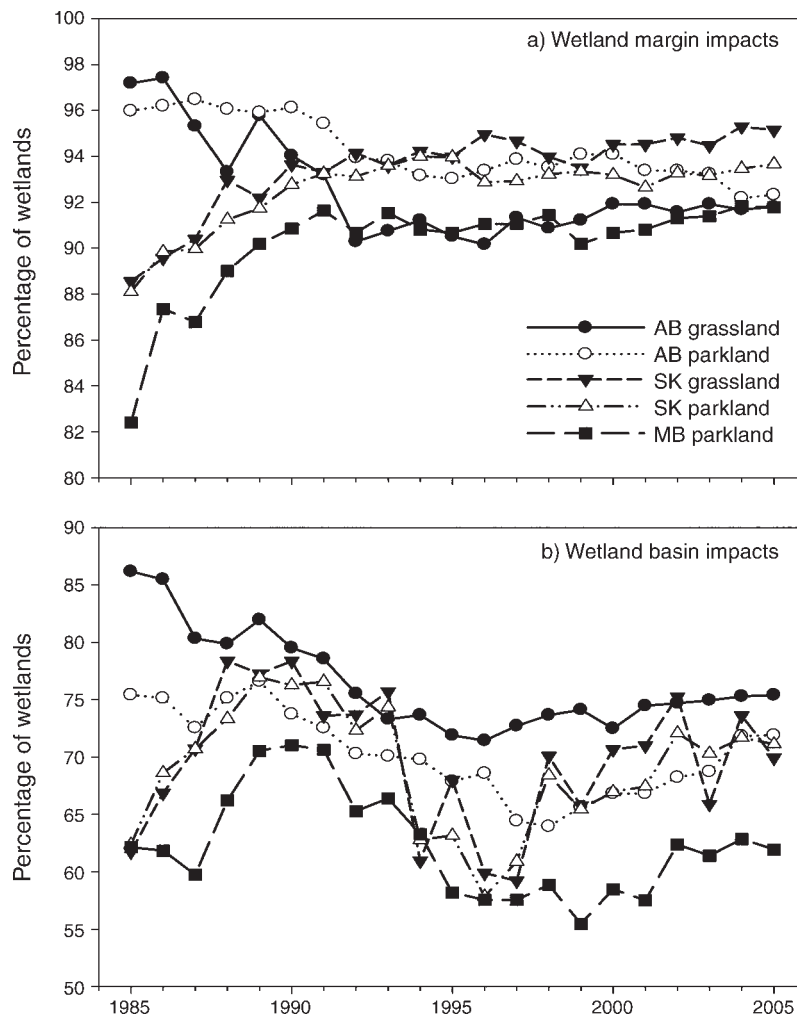


FIG. 7. Percentage of wetlands across the Canadian prairie pothole region with impacts (1985–2005): (a) percentage of sample wetlands with impacts to margins; (b) percentage of sample with impacts to basins.

was Alberta grassland, where recovery rates exhibited a slight decline with increasing pond density. In Saskatchewan, slopes for impact and recovery rates for grassland were higher than for parkland, suggesting that basin impacts in grassland are more sensitive to changes in pond density. In Alberta, impact rates for grassland also had a steeper slope than parkland, but these results should be considered in light of the anomalous recovery rates in Alberta grassland as mentioned. As such, the data do not conclusively indicate that transition rates fluctuate more in grassland than in parkland regions, although some evidence is consistent with this possibility.

Patterns in transition rates relative to wetland type also generally conformed to predictions, with ephemeral and seasonal wetlands having higher impact rates. Margins of pristine streams and seasonal, permanent, and artificial wetlands all had similar impact rates. Because margins are less prone to flooding than basins, they are likely subject to the same risk of agricultural impact regardless of wetland permanence. Pristine ephemeral wetlands had the lowest rate of impact to wetland margins. However, pristine ephemeral wetlands had the highest rate of impact when both the margin and basin were affected, likely because they are shallow and most prone to dry out completely in a single year. Pristine and partially impacted wetlands with higher degrees of permanence were less likely to transition to fully impacted states. Pristine artificial wetlands had the lowest rates of impact to fully impacted states, likely because deep basins, such as dugouts and borrow-pits, are last on the landscape to go dry during periods of drought. Rates of recovery to pristine states were equally low for all water bodies, which is consistent with the high incidence of wetland impact observed throughout the study period.

As expected, impact rates were generally highest for wetlands in cultivated and pasture lands while recovery rates were highest for wetlands in more natural grass and wooded areas. Exceptions were transition parameters between partially and fully impacted wetlands where impact and recovery rates were almost equal among land-use types. Impact rates in wooded and grassed areas may be lower because the land is agriculturally marginal and less profitable due to the added expense of conversion to "usable" land. Grassed lands may also include areas under conservation covenants, where the land is to be managed as "native" prairie. Wetland recovery in natural areas may occur more quickly because these areas are not used for agriculture on an annual basis. In contrast, pasture and cultivated lands are routinely grazed and farmed, respectively, so wetlands in these areas have both high impact rates and less opportunity to recover. Because transition parameters between partially and fully impacted wetlands varied with pond density, land-use differences between impact and recovery rates should also vary with pond density. Saskatchewan parkland was relatively wet

in 1995 (~ 12 ponds/km²), resulting in recovery rates at their highest levels and impact rates at their lowest levels. If we had chosen a drier year as our standard for land-use comparisons, impact rates may have been higher than recovery rates for most land uses.

A potential criticism of our methods is that land use and wetland type were modeled as additive effects in the most parsimonious models, and no attempt was made to include these individual covariates as multiplicative effects. However, we had no a priori expectation that trends among land uses or wetland types might vary spatially, and such complex relationships would be difficult to interpret. Another possible limitation was assigning a single dominant land use to wetlands when land uses can vary over time. On average, 8% of sample wetlands per year were surrounded by a land use that differed from their designated dominant land use. A possible resolution may be to model land uses as time-varying individual covariates. However, that approach in this case study would have been impractical and difficult computationally. Transition estimates of wetlands with different land uses were generally consistent with expectations, which led us to believe the estimates from dominant land uses were defensible.

Although impact detection consistency between observers was high for basin and margin impacts in 2007, consistencies of 91% and 93% in 2006 for basin and margin impacts, respectively, may be considered low. However, because observers are arbitrarily assigned to survey wetlands, these types of discrepancies were unlikely biased toward particular wetlands in the data set. Therefore, the observed level of misclassification of wetland states would not explain the directional trends exhibited in the transition rate estimates.

Implications for management

In response to record low duck populations and intensified agriculture, the North American Waterfowl Management Plan was established in 1986 to increase duck populations to average levels of the 1970s (U.S. Fish and Wildlife Service and Canadian Wildlife Service 1986). To achieve this goal, the plan placed priority on preventing further habitat loss and degradation, and on restoring habitat. To implement North American Waterfowl Management Plan strategies in prairie Canada, Prairie Habitat Joint Venture was created and since its inception has secured over 1.9 million ha of nesting and wetland habitat (North American Waterfowl Management Plan; *available online*).⁵ Activities on these lands include enhancement, restoration, preservation, and land stewardship. Despite these conservation activities, the present study demonstrates that impacts to wetlands across the region continue.

If we had only examined impact rates on pristine wetlands, we might have inferred conservation efforts

⁵ <http://www.nawmp.ca/pdf/HabMat2007e.pdf>

were effective because rates decreased over time. However, when impact rates were examined in conjunction with recovery rates and overall incidence of wetland impacts, declining impact rates were evidently a function of the decreasing number of pristine wetlands remaining on the landscape. More than 90% of wetlands had some form of visible agricultural impact. Our findings were rendered from a single modeling framework in Program MARK using multistate models. To our knowledge, this is the first study to apply these models for quantifying wetland impact and recovery rates. We are unaware of any other technique that could simultaneously estimate all transition probabilities and enable the examination of how these probabilities covary with multiple factors and over time. Multistate models may also have utility in estimating rates of other dynamic landscape processes, such as land reclamation and restoration from natural and anthropogenic disturbances.

Our estimates of recovery rates for wetlands may be high because wetland function was not quantified. For instance, reflooded wetlands that appear to have recovered may not be functionally equivalent to a truly pristine wetland. Consideration of other inputs, such as sedimentation or agricultural fertilizers and pesticides, would undoubtedly have increased the percentage of wetlands with impacts from the estimates determined in this study (Houlahan and Findlay 2004, Voldseth et al. 2007). The modeling framework we employed in the present study could also be used to measure recovery and impact rates from such inputs in response to changes in policies or other intervention.

Political, economic, and other environmental factors may contribute to spatial variation in wetland impact and recovery rates; model comparisons in the present study demonstrated differences resulting from more than variation in overall density of wetland basins between regions. Fluctuating commodity prices for agricultural products and agricultural subsidies may influence rates of wetland impact (Bradshaw and Smit 1997, Drever et al. 2007). Wetland size and distance to roads could also influence impact and recovery (Douglas and Johnson 1994, Daniels and Cumming 2008). These factors could be considered as covariates in future analyses. Moreover, we defined transition states based on agricultural impacts, but wetland states could be defined based on other criteria such as vegetative community structures, water nutrient and sedimentation levels, or invertebrate abundance (Dobkin et al. 1998, Gleason et al. 2003, Houlahan and Findlay 2004). If wetland destruction events had occurred frequently in the data set, we might have also estimated wetland "survival" (White et al. 2006).

Conservation recommendations based on our results include placing priority on preserving remaining pristine wetlands due to both the high incidence of impact and higher rates of impact than recovery. Priority should also be given to ephemeral and seasonal wetlands, which generally had higher impact and lower recovery rates

than wetlands with greater permanency (Whigham 1999). Adjacent land uses should also be considered in wetland conservation planning (Daniels and Cumming 2008); efforts could focus on wetlands in cultivated and pasture lands as they are at highest risk of impact. In addition, basin impact rates will likely increase and recovery rates will decline in the future if prairie climate becomes progressively drier as projected (Johnson et al. 2005). Because we identified wetlands at the greatest risk of impact and estimated effects of putative factors on wetland impact and recovery, the results of this study have further relevance when considering conservation of wetland-associated species.

In addition to documentation of trends in wetland impacts for prairie Canada, this study is also unique to the extent that we simultaneously estimated the contributions of particular factors on wetland impact and recovery. Many studies document patterns, but few attempt to simultaneously elucidate the drivers of observed patterns. Clearly, wetland impact is a prairie-wide problem, and stronger wetland protection is needed for all three prairie provinces. Because precipitation cannot be controlled for, conservation efforts should focus on land-use practices. In particular, attempts should be made to mitigate grazing and cultivation practices, as the majority of wetlands are surrounded by these types of land uses and impact rates are highest in these areas. To ensure future management and conservation efforts are effective, we urge continued monitoring of agricultural impact and recovery of wetlands in prairie Canada.

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